

Two cliffs, not one: bidirectional nutrient targeting, the supplement blind spot, and the limits of current diet-quality assessment

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Running title. Bidirectional nutrient targeting

Keywords. diet quality; dietary reference intakes; estimated average requirement; dietary supplements; nutrient adequacy

Abbreviations. AI, Adequate Intake; AR, Average Requirement (European Food Safety Authority); DRI, Dietary Reference Intake; EAR, Estimated Average Requirement; EFSA, European Food Safety Authority; EVM, Expert Group on Vitamins and Minerals (United Kingdom); LRNI, Lower Reference Nutrient Intake (United Kingdom); NASEM, National Academies of Sciences, Engineering, and Medicine; NHANES, National Health and Nutrition Examination Survey; PRI, Population Reference Intake (EFSA); RDA, Recommended Dietary Allowance; RNI, Reference Nutrient Intake (United Kingdom); UL, Tolerable Upper Intake Level.

Abstract

Objectives. To examine whether diet-quality assessment represents the two-sided nature of nutrient–health relationships; to determine which reference value licenses an individual inference; to identify how intakes above a limit are reached; and to propose a two-sided component score for use within an existing index.

Background. Risk rises at both ends of the intake range for many micronutrients, yet diet-quality indices score components monotonically and exclude supplements by construction, while population reference values are used as individual targets despite guidance against this.

Methods. Critical narrative review. Literature was identified through PubMed/MEDLINE, Scopus and the Cochrane Library to September 2026, with direct consultation of the reference-value documents of the National Academies of Sciences, Engineering, and Medicine, the relevant United Kingdom expert committees, and the European Food Safety Authority. Selection was purposive rather than systematic.

Results. Seven vitamins have no upper limit because adverse-effect data are insufficient, not because safety is shown, yet effects of excess are documented for most micronutrients, differing in study design, not in whether harm occurs. Where limits exist, that gap is three- to seven-fold for vitamin A, zinc, iron and selenium. Supplements supply one to three orders of magnitude above requirement and drive most exceedance in adults; fortification does in children. No analysis reports aggregate exceedance across nutrients, and solubility does not predict safety. Of eight instruments in routine use, none applies a two-sided rule at nutrient

level or counts supplements; a smaller nutrient-based literature does more, but none does all. We set out a candidate module covering every micronutrient, anchored on the estimated average requirement and the upper limit, separating food, fortification and supplements, and weighting penalties across four tiers.

Conclusion. Two-sided, supplement-aware assessment is constructible now. We set out the staged descriptive, biochemical and prognostic work that would establish whether it is worth adopting.

1. INTRODUCTION

Nutritional adequacy is usually discussed as though it had one direction. Public health messaging, dietary guidance and the indices built to operationalise them are organised around the idea of getting enough — enough fibre, enough calcium, enough vitamin D — and the scoring systems that quantify diet quality largely reflect that orientation. For a substantial number of nutrients, however, the relationship between habitual intake and risk is U-shaped or J-shaped: harm accumulates at low intakes, falls across a range of adequacy, and rises again as intake climbs. Iodine is the textbook case, with both deficiency and excess producing thyroid dysfunction.^{1,2} Selenium, iron, zinc, vitamin A, vitamin B6 and vitamin E all have well-described upper-end toxicities.^{3–7} The framework that defines nutrient requirements has acknowledged this since the tolerable upper intake level was formalised in the late 1990s.⁸

Yet the instruments most widely used to score diet quality are, with limited exceptions, monotone in their components. Higher scores are awarded for more of the things considered beneficial and fewer of the things considered harmful, and in the great majority of cases no scoring penalty attaches to an intake that has climbed past the point of benefit. This is not an oversight so much as an inheritance: the indices were built to discriminate between broadly inadequate and broadly adequate diets in populations where inadequacy was the dominant problem, and they perform that task well.^{9,10} The dietary-pattern evidence they encode is itself substantial, including randomised evidence for pattern-level change.¹¹ Whether they remain fit for purpose in populations where fortification, supplementation and abundant food supply have made the upper end a live concern is a different question, and one that has attracted less attention than it deserves.

This review makes three linked arguments. The first is that bidirectional risk is real, is well characterised for a subset of nutrients, and is largely unrepresented in diet-quality scoring. The second is that the reference values used to judge whether an individual's intake is adequate are frequently the wrong ones: the recommended dietary allowance and its international equivalents are population reference values, and the guidance that accompanies them says explicitly that they should not be used to assess an individual's intake. Yet that is how they are most commonly used, in clinical practice, in consumer applications and in the construction of personal intake targets.^{12,13} The third is that intakes above an upper limit are reached by a specific and identifiable route, principally concentrated supplements and fortified staples, and only rarely through whole foods, and that no widely used assessment instrument counts either of them in a way that would allow the upper end of the distribution to be seen at all. Having set out these three, we propose what a two-sided component score would require.

These three arguments share a structure. In each case a tool that is well suited to one inferential task (describing populations, ranking diets, screening for inadequacy in resource-poor settings) has been carried across to a different task, that of characterising an individual, without the accompanying change in what may legitimately be concluded. Our aim is not to disparage the instruments, all of which do what they were designed to do, but to be precise

about what they license and what they do not, and to set out what a bidirectional, individually interpretable assessment would require.

2. METHODS

This is a critical narrative review. It was not conducted as a systematic review, was not registered, and does not claim exhaustive coverage of the literatures it draws on. It is reported in accordance with SANRA, the Scale for the Assessment of Narrative Review Articles, which asks that a narrative review justify its importance, state its aims, describe how the literature was identified, reference appropriately, and present evidence and data so that the reader can judge the strength of each claim.¹⁴

Literature was identified through searches of PubMed/MEDLINE, Scopus and the Cochrane Library from database inception to September 2026, using combinations of the terms diet quality index, nutrient adequacy, dietary reference intakes, estimated average requirement, tolerable upper intake level, nutrient excess, dietary supplements, fortification, U-shaped, non-monotonic and nutrient toxicity. Reference lists of retrieved articles were hand-searched, and the primary reference-value documents of the National Academies of Sciences, Engineering, and Medicine, the UK Committee on Medical Aspects of Food Policy and its successor the Scientific Advisory Committee on Nutrition, the UK Expert Group on Vitamins and Minerals, and the European Food Safety Authority were consulted directly and not through secondary summaries.

A targeted search was also made for prior instruments of the kind proposed in Section 8. Alongside the database searches described above, we examined the dietary guidance and reference-value documents of the United States Departments of Agriculture and of Health and Human Services, the European Food Safety Authority and the United Kingdom expert committees, together with the defining publication of each diet-quality instrument examined, for any scoring scheme applying a two-sided rule at the level of an individual nutrient or counting supplemental intake. Neither the national dietary guidance of the three jurisdictions nor any of those instruments does either; a smaller family of nutrient-based indices that does is described in Section 6. The search was purposive, not systematic, and included no controlled-vocabulary or patent search, so statements here about the absence of a prior instrument record what we located rather than what exists.

Sources were selected purposively to support a critical argument and to represent competing positions where the evidence is contested, particularly on the upper limbs of intake–risk curves, where the literature is thinner and the trial evidence is mostly of supplementation, not of habitual dietary intake. Selection and appraisal were performed by the authors without duplicate independent screening. The same applies to the assignment of nutrients to evidence tiers in Section 8.4: it is a structured summary of the authors' reading of the located literature, not a formal evidence appraisal, and it has not been replicated by an independent reader. The review is therefore subject to selection bias, and readers should treat the coverage of each individual nutrient as illustrative and not as complete. Where we describe the properties of a published index, we cite the index's own defining publication and have verified the scoring description against it.

3. THE REFERENCE VALUES, AND WHAT EACH ONE LICENSES

3.1 The guidance, and the practice

The Dietary Reference Intakes comprise four values, and the report devoted to their application in dietary assessment is unambiguous about which of them supports which

inference.¹² The estimated average requirement (EAR) is the median requirement of a group: half of a comparable population requires more, half less. The recommended dietary allowance (RDA) is derived from the EAR and set roughly two standard deviations above it, so as to meet the requirements of approximately 97.5% of the population. The adequate intake (AI) is used where the evidence was insufficient to derive an EAR at all. The tolerable upper intake level (UL) marks the highest habitual intake expected to pose no risk of adverse effects. We refer to this value as the upper limit throughout, and use UL only where brevity requires it; its counterparts in the United Kingdom and European systems are named in Figure 1.

The consequence, stated explicitly in the guidance and repeated in the subsequent applications literature, is that the RDA should not be used to assess the adequacy of an individual's intake.^{12, 13} An intake below the RDA cannot be called inadequate, because the RDA exceeds the requirement of all but a small fraction of people; a person consuming below it may comfortably exceed their own requirement. The EAR is the value against which an individual's usual intake should be compared, and the comparison yields a probability rather than a verdict. Where only an AI exists, an intake above it suggests low probability of inadequacy, but an intake below it supports no quantitative conclusion whatever — a point that is regularly lost in translation.

Practice diverges from this guidance in a specific and consequential way. The RDA is the value that appears on food labels, in consumer applications, in supplement marketing and in most personal target-setting tools, and it is routinely presented as a threshold that individuals should reach. The inferential error is not that the RDA is a bad number; it is that a value constructed to sit above almost everyone's requirement is being used as though it were each person's requirement. The result systematically overstates the prevalence of individual inadequacy and, where it drives supplementation, pushes intakes toward the upper limb of a curve whose position is often unknown.

The distinction between group and individual assessment sharpens the point. For a group, the prevalence of inadequate intake can be estimated by the EAR cut-point method, in which the proportion of the group whose usual intake falls below the EAR approximates the proportion whose intake is inadequate, provided intakes and requirements are approximately independent and the requirement distribution is roughly symmetric.^{13, 15} Where those conditions fail, as they do for iron in menstruating women because the requirement distribution is markedly skewed, a full probability approach is required instead. For an individual, neither method returns a verdict: the output is a probability that the person's usual intake is below their own unobserved requirement, and that probability is bounded away from zero and one across most of the plausible intake range.

What follows is a claim about the honesty of outputs; the arithmetic is not in dispute. An assessment tool that reports “you are not meeting your requirement for magnesium” has asserted something it cannot know. One that reports “your usual intake is below the median requirement, so there is a moderate probability that it is inadequate for you” has said what the data support. The second formulation is less satisfying and considerably more defensible, and the gap between the two is where a large amount of consumer nutrition advice currently operates.

3.2 The same distinction in the UK and European systems

This is not a peculiarity of the North American framework. The UK dietary reference values, established by the Committee on Medical Aspects of Food Policy and maintained by the Scientific Advisory Committee on Nutrition, use an EAR with the same definition, a

reference nutrient intake (RNI) that occupies the position of the RDA, and a lower reference nutrient intake (LRNI) set approximately two standard deviations below the EAR and used in national surveillance to flag intakes very likely to be inadequate. The European Food Safety Authority uses an average requirement (AR) and a population reference intake (PRI) with the same respective meanings, together with an adequate intake and a tolerable upper intake level; its nutrient-specific opinions state the derivation explicitly, and its treatment of iron illustrates the skewed-requirement problem discussed above.¹⁶

Figure 1 aligns the three systems. The alignment matters for two reasons. It shows that the inferential rule is international, not jurisdictional — in every system, the median-requirement value is the one that supports an individual-level statement. And it exposes a translation hazard: the RNI and the PRI are equivalent to the RDA and not to the EAR, so a target sheet built in one jurisdiction and applied in another inherits the same error in a different vocabulary. Table 1 sets out what each value does and does not license.

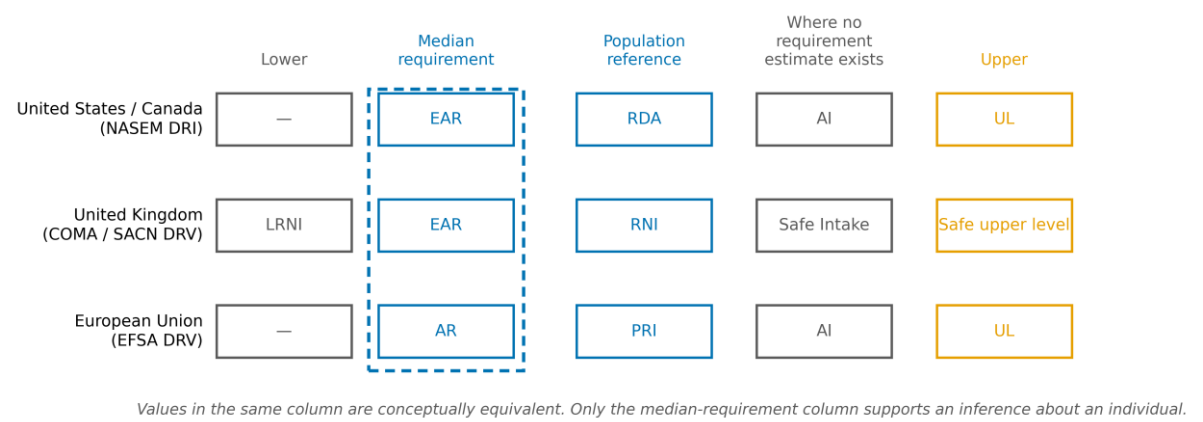


Figure 1. Alignment of reference values across the United States and Canada, the United Kingdom and the European Union. Values in the same column are conceptually equivalent. The median-requirement column, outlined, contains the only values that support a probabilistic inference about an individual’s intake; the population-reference column is frequently but incorrectly used for that purpose.

Reference value	What it is	Valid use for an individual	Common misuse
EAR (US/UK) · AR (EFSA)	Median requirement of a comparable group	Compare usual intake against it to estimate the probability that intake is inadequate	Rarely misused — more often simply unused
RDA (US) · RNI (UK) · PRI (EFSA)	Set ~2 SD above the EAR; meets the requirements of ~97.5% of the population	None. It is a population reference value	Treated as a personal target; intakes below it labelled deficient
AI	Used where evidence was insufficient to derive an EAR	Intake above it suggests low probability of inadequacy	Intake below it treated as inadequate, which it does not support
UL	Highest habitual intake expected to carry no risk of adverse effects	Compare usual intake against it to flag potential excess	Absence of a UL read as evidence of safety
LRNI (UK only)	~2 SD below the EAR; meets the needs of ~2.5% of the population	Intake below it is very likely inadequate	Used as a target rather than as a floor

Table 1 What each reference value licenses. The pattern is consistent across the three systems: only the median-requirement value supports a probabilistic statement about an individual.

3.3 Measurement error compounds the problem

Even the correct comparison is fragile in practice, because the quantity being compared is measured badly. Biomarker-based validation studies have shown substantial and systematically biased under-reporting of energy and protein intake by both food-frequency questionnaires and 24-hour recalls, with the degree of misreporting related to body mass index and other participant characteristics.^{17,18} Because usual intake must be estimated from a small number of observation days, within-person variability further widens the uncertainty around any individual estimate.¹⁹ An individual-level adequacy statement therefore carries two distinct sources of error: the requirement is unknown, and the intake is imprecisely measured. Honest reporting would attach an interval to it and not a point estimate.

4. THE TWO-SIDED INTAKE–RISK RELATIONSHIP

4.1 The shape of the curve, and what the reference values mark on it

The canonical representation of nutrient risk, formalised in the risk-assessment model the National Academies adopted for setting upper limits, plots probability of an adverse outcome against habitual intake and shows two rising limbs separated by a range of intakes at which neither deficiency nor excess is expected.⁸ Figure 2 reproduces this structure and marks the reference values on it.

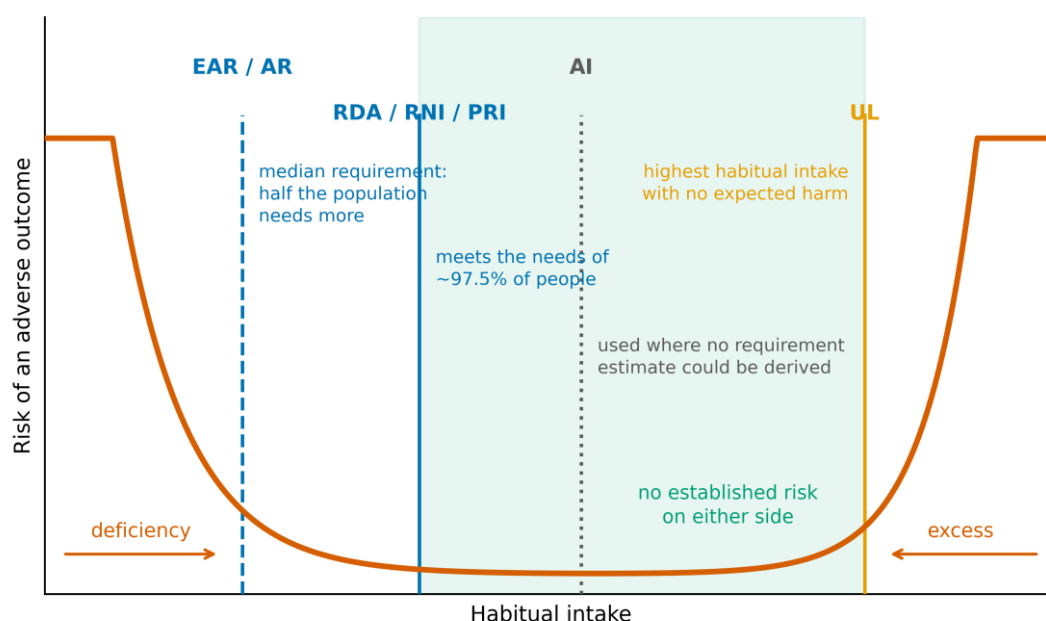


Figure 2. The two-sided intake–risk relationship and the reference values that mark it. Risk rises at both low and high habitual intake, separated by a range in which neither is expected. The lower limb carries two values derived from a requirement distribution (EAR or AR; RDA, RNI or PRI); the upper limb carries one derived by risk assessment (UL). The adequate intake (AI) is used where no requirement estimate could be derived. The distance between lower and upper values differs markedly by nutrient and is not to scale.

Three features of that figure carry unusual weight. The reference values are not symmetric: the lower limb is populated by two values derived from requirement distributions, while the upper limb is marked by a single value derived from a no-observed-adverse-effect level and an uncertainty factor, and the two are not commensurable. The distance between them varies enormously by nutrient, from wide for vitamin C or thiamin to narrow for iodine, selenium and preformed vitamin A, so that the practical difficulty of staying between the cliffs is specific to the nutrient and does not generalise. And for a substantial number of nutrients no

upper level has been set at all, which reflects absence of evidence rather than evidence of safety.

The deficiency limb nevertheless remains the more important of the two at global scale. Dietary risks were estimated to account for eleven million deaths worldwide in 2017, the largest attributable burden of any single risk factor assessed in that analysis, and the great majority of that burden reflects inadequate intake of protective foods, not excess of micronutrients.²⁰ Nothing in this review should be read as suggesting that the upper limb deserves equal weight in global nutrition policy. The argument is narrower: in populations where fortification and supplementation are widespread, the upper limb becomes relevant for a non-trivial minority of individuals, and the instruments used to assess those individuals cannot currently see it.

4.2 Where the upper limb is well characterised

For a minority of nutrients, the excess side is documented with a confidence approaching that of the deficiency side. Iodine intake shows a well-described U-shaped relationship with thyroid dysfunction, with both insufficiency and excess implicated in hypothyroidism and in autoimmune thyroiditis.^{1, 2} Selenium has a comparably narrow window: status below approximately 100 µg/L is associated with adverse outcomes, while supplementation in already-replete populations has been associated with an increased incidence of type 2 diabetes.^{3, 4} Pyridoxine at high supplemental doses produces a sensory neuropathy that was described in detail four decades ago and continues to be reported.⁵ Iron overload, whether genetic or iatrogenic, produces well-characterised organ damage.⁶ Chronic high-dose zinc induces copper deficiency.⁷

Two trial results deserve particular emphasis because they illustrate that the upper limb can be steep enough to reverse the direction of a public-health recommendation. Beta-carotene supplementation, undertaken on the strength of consistent observational associations between carotenoid intake and lower lung-cancer risk, increased lung-cancer incidence in smokers in two independent trials.^{21, 22} A meta-analysis of high-dose vitamin E supplementation trials reported a dose-dependent relationship with all-cause mortality.²³ Whatever the residual controversy about the pooled estimates, the episode establishes the principle: for these nutrients, the assumption that more is better was not merely unsupported but wrong.

4.3 Two structural features of the limits themselves

The first is coverage: for a substantial minority of nutrients no upper limit exists at all. In the North American system a numeric UL has been set for eight vitamins (vitamin A, vitamin C, vitamin D, vitamin E, niacin, vitamin B6, folate and choline) and for around fifteen minerals and trace elements, while seven vitamins are formally recorded as not determinable: vitamin K, thiamin, riboflavin, vitamin B12, pantothenic acid, biotin and the carotenoids.¹⁵ Those entries are frequently misread. They do not certify safety. The National Academies record them as not determinable "due to lack of data of adverse effects", and caution explicitly that this "does not mean that there is no potential for adverse effects resulting from high intakes".¹⁵ Absence of a limit is absence of evidence.

The second is width. Where both a requirement and a limit exist, the arithmetic distance between them is narrower than the phrase "upper limit" tends to suggest. Taking the published adult values and dividing the limit by the recommended intake gives, for men: vitamin A roughly 3.3-fold (900 µg retinol activity equivalents against a limit of 3,000 µg of preformed retinol), zinc 3.6-fold (11 mg against 40 mg), iron 5.6-fold (8 mg against 45 mg), selenium 7.3-fold (55 µg against 400 µg), and iodine 7.3-fold (150 µg against 1,100 µg).²⁴

Vitamin C is more forgiving at roughly 22-fold (90 mg against 2,000 mg), and calcium narrower still at 2.5-fold (1,000 mg against 2,500 mg). These are not uniformly generous margins, and for several nutrients a single well-marketed supplement can traverse the entire window in one tablet.

Magnesium is an instructive anomaly, exposing how easily these values are misread. Its recommended intake for men is 400–420 mg while its upper limit is 350 mg, so that the ceiling appears to sit beneath the floor. The resolution is that the two values do not refer to the same quantity: the magnesium limit "represent[s] intake from a pharmacological agent only and do[es] not include intake from food and water".²⁵ The same scope-dependence applies elsewhere. The folate limit applies to synthetic folic acid from fortified food and supplements but not to food folate; the vitamin A limit applies to preformed retinol only, not to provitamin A carotenoids; the niacin and vitamin E limits likewise apply to synthetic or supplemental forms.^{24, 26} A ratio computed without regard to scope is meaningless, and any index that compares intake against a limit must know which intake the limit was written about.

Upper limits are also less uniform in their evidentiary basis than their presentation suggests. They are derived by dividing a no-observed-adverse-effect or lowest-observed-adverse-effect level by a composite uncertainty factor that has ranged from 1 to 36 across nutrients.¹² Renwick's appraisal of micronutrient toxicology is blunt about the consequence: limited data exist on the toxicity of most micronutrients, few studies meet the criteria applied to other food chemicals, and there is no clear scientific rationale for the reduced uncertainty factors used in this field.²⁷ Jurisdictions consequently disagree on the same nutrient. The selenium limit is 400 µg in the United States but 255 µg in the European Union, and the vitamin E limit 1,000 mg against 300 mg.^{28, 29} A bidirectional index should therefore weight its upper-side penalties by the strength of the evidence behind each limit instead of treating every limit as an equally firm edge.

4.4 Which nutrients qualify today

Table 2 summarises the nutrients for which both limbs are defined well enough to support two-sided scoring today. It is a short list, and its shortness is the point.

Nutrient	Deficiency-side outcome	Excess-side outcome	Basis for the upper limb
Iodine	Goitre, hypothyroidism, impaired neurodevelopment	Hypothyroidism, autoimmune thyroiditis, hyperthyroidism	Population studies across the intake range; UL defined ^{1, 2}
Selenium	Keshan disease; impaired immune and thyroid function	Selenosis; raised incidence of type 2 diabetes in replete populations	Randomised supplementation trial; UL defined ^{3, 4}
Vitamin A (preformed)	Xerophthalmia, impaired immunity	Hepatotoxicity, teratogenicity, reduced bone density	Clinical and observational; UL defined ⁸
Vitamin B6	Peripheral neuropathy, dermatitis	Sensory neuropathy at high supplemental doses	Case series and clinical description; UL defined ⁵
Iron	Anaemia, impaired cognition	Organ damage from overload	Clinical; UL defined ⁶
Zinc	Impaired growth and immunity	Copper deficiency, impaired immune function	Controlled feeding studies; UL defined ⁷

Nutrient	Deficiency-side outcome	Excess-side outcome	Basis for the upper limb
Vitamin E	Rare; neurological in malabsorption	Dose-dependent association with all-cause mortality in trials	Meta-analysis of supplementation trials ²³
Beta-carotene	Contributes to vitamin A status	Increased lung-cancer incidence in smokers	Two randomised trials ^{21, 22}

Table 2 Nutrients for which both limbs of the intake–risk relationship are characterised well enough to support two-sided scoring. Most nutrients with a defined upper limit do not appear here, because a defined limit is not the same as a documented outcome.

The distinction drawn in the final column matters. A tolerable upper intake level is a risk-assessment construct, derived by applying uncertainty factors to a no-observed-adverse-effect level, and its existence does not imply that adverse outcomes have been observed at intakes just above it in free-living populations.⁸ For several nutrients the limit rests on a single study in a small sample. Treating every limit as an equally firm cliff edge would overstate what is known, and a bidirectional index should weight its upper-side penalties according to the strength of the underlying evidence instead of applying a uniform rule.

It is also worth correcting an impression that the short list above may create. Documented effects of excess are not confined to a handful of nutrients. Assembling the excess-side literature for every micronutrient carrying a reference value yields described adverse effects for the large majority of them; what differs between nutrients is not whether harm has been reported but the design of the studies reporting it. Section 8 grades that evidence nutrient by nutrient and uses the grading to set the depth of the scoring penalty.

4.5 Where it is not — and why that matters for index design

There is a further asymmetry in how the two sides of the curve have been studied, and it is visible in the structure of the literature and not in any single study. Deficiency has been aggregated across nutrients: one analysis of a national survey reports that 31% of the United States population is at risk of at least one vitamin deficiency or anaemia, with 6.3% at risk in two nutrients and 1.7% in three to five.³⁰ No equivalent analysis exists on the excess side. No published estimate exists, for any general population, of the proportion exceeding the upper limit for one or more nutrients considered together. The question has been asked systematically of deficiency and not of excess.

The two sides also differ in the character of their evidence. Nutrient toxicities tend to be documented as recognisable clinical entities reported in case series, not in cohorts or trials: zinc-induced copper deficiency, with its combination of myeloneuropathy and cytopenias, is described almost entirely in this way.³¹ Such evidence establishes that a harm exists but generalises poorly to dose–response, which is precisely what is needed to place a threshold. Deficiency, by contrast, has accumulated a large secondary and interventional literature. The consequence is not that excess-side evidence is of poor quality but that it is differently shaped, and less suited to the quantitative task of drawing a curve.

No bibliometric analysis quantifies this contrast, so we make the point qualitatively and not numerically. It is nonetheless recognised within the bodies that set the values. Reviews of upper-level methodology note the limited confidence attaching to individual limits and the difficulty of grading excess-risk evidence consistently across nutrients.³²

This asymmetry has a direct implication for index construction, and it limits our own argument. An index cannot score bidirectionally for a nutrient whose upper limb has not been located. Any bidirectional scoring scheme will therefore be partial, applying two-sided rules

to the minority of nutrients for which upper thresholds are defensible and one-sided rules elsewhere. That is a weaker proposal than a fully symmetric index, and it is the strongest proposal the evidence currently supports.

4.6 Two nutrients can interact across their limbs

A further complication, and one that no current index represents, is that the upper limb of one nutrient can interact with the deficiency state of another. The most discussed instance concerns folic acid and vitamin B12. High folic acid intake can correct the anaemia of B12 deficiency while the neurological damage progresses, and the concern that fortification might therefore mask B12 deficiency in older adults has shaped fortification policy debates for two decades.³³ The evidence remains contested — the interaction is better established as a biochemical phenomenon than as a population-level harm, and observational findings linking high folate with low B12 status to cognitive outcomes have been difficult to replicate consistently.³⁴ We cite it not as settled but as an illustration of a structural point: a scoring system that treats each nutrient as an independent component cannot represent a risk that arises from the joint configuration of two.

4.7 Where the two-sided framing is genuinely contested

For two of the most prominent candidate U-shapes, the shape itself is disputed. Sodium is the clearest case. Observational analyses using morning fasting urine estimates have reported a J-shaped association with cardiovascular outcomes, with the lowest risk at intakes above those recommended by guidelines.³⁵ Critics have argued that the finding is an artefact of reverse causation and of systematic error in estimating usual sodium intake from spot samples, and the trial evidence on blood pressure continues to support reduction. The disagreement is methodological and not merely interpretive, and it has not been resolved.

Vitamin D is the second. Non-linear Mendelian randomisation has been reported to support a causal contribution of low status to dementia risk,³⁶ while large supplementation trials have not shown the corresponding benefits for major clinical outcomes,³⁷ and reanalyses using revised non-linear methods have weakened the earlier causal claims. A bidirectional index would have to take a position on both nutrients, and neither position could currently be described as evidence-based in the strong sense. Alcohol offers a third instance, where an apparent protective effect at low intake has been challenged by burden-of-disease and Mendelian randomisation analyses concluding that the intake minimising health loss is zero.³⁸ Our proposal is deliberately restricted to nutrients where the direction of the upper limb is not in dispute; extending it to sodium, vitamin D or alcohol would import a controversy rather than resolve one.

5. THE SUPPLEMENT CHANNEL

Everything said so far concerns the shape of the intake–risk relationship. It leaves open a practical question that determines whether any of it matters: by what route does an ordinary person in a high-income country actually arrive at an intake near the upper limb? Analyses that decompose usual intake into naturally occurring food, enrichment and fortification, and supplements give a reasonably clear answer for adults, and a more complicated one for everyone else.^{39–41}

For most nutrients, and for the population aged two years and over, supplements are the largest incremental contributor to intakes above a limit. Adding them to food and fortification raises the proportion above the limit from essentially zero to around 2% for iron, 2.6% for calcium, and 0.5% for vitamin B6 and vitamin D.⁴² For three nutrients, however, fortification

alone already produces material exceedance before any supplement is taken: in the same population, 8.8% exceed the folic acid limit and 1.9% the niacin limit from enriched food alone, and 4.6% exceed the zinc limit from naturally occurring food by itself.⁴²

In young children the ordering reverses. Among those aged two to eight, food and fortification alone place 41.7% above the folate limit, 39.9% above the zinc limit and 10.1% above the niacin limit; supplements raise these further, to 52.4%, 46.3% and 18.5%, but they are the minority contributor.⁴² Whether this constitutes a health problem is a separate question — the paediatric limits for zinc, vitamin A and folate are extrapolated from adult data with large uncertainty factors,⁴⁰ and several of the investigators who report these figures argue the limits themselves may be set too low.⁴³

Nor are whole diets uniformly self-limiting. For a small set of nutrient–food combinations an ordinary portion exceeds the limit outright: a single serving of liver supplies preformed vitamin A at several times the adult limit; five or six Brazil nuts approach or exceed the selenium limit; dried seaweed can deliver iodine at many multiples of it; and a modest serving of oysters can meet or exceed the zinc limit. Documented toxicity has followed each of these.⁴⁰ A heavy enriched-grain intake, the fifteen to twenty slices of bread that a large eater might consume, can likewise bring niacin toward its limit from food alone, which is consistent with fortification, not supplementation, being the operative route for that nutrient.

One inference that might seem natural does not survive contact with the evidence. Since around 72% of United States adults are overweight or obese (crude prevalence 72.4%),⁴⁴ and micronutrient intake rises with energy intake, one might expect high-energy eaters to exceed limits from food alone. The literature points the other way. Energy-dense diets are characteristically micronutrient-dilute, and people with obesity show higher rates of micronutrient inadequacy and not of excess.^{45–47} No published analysis has stratified food-only exceedance by energy intake and found it rising. Eating a great deal is not, on the current evidence, a route to the upper limb; it is more often a route to the lower one.

5.1 The doses are not marginal

The first thing to establish is the scale of the discrepancy between what physiology requires and what the market supplies, because it is larger than most discussions of nutrient excess assume. Vitamin B12 is the clearest case. The adult requirement is 2.4 µg per day. Retail preparations are sold at 500, 1,000 and 5,000 µg per tablet, between roughly 200 and 2,000 times requirement. Biotin, with an adequate intake of 30 µg, is marketed at 5,000 to 20,000 µg for hair and nail indications, some 170 to 670 times requirement, and is used in multiple sclerosis research at 300 mg, approximately ten thousand times. Figure 3 sets out the comparison across nutrients on a logarithmic scale, which is the only scale on which it can be drawn.

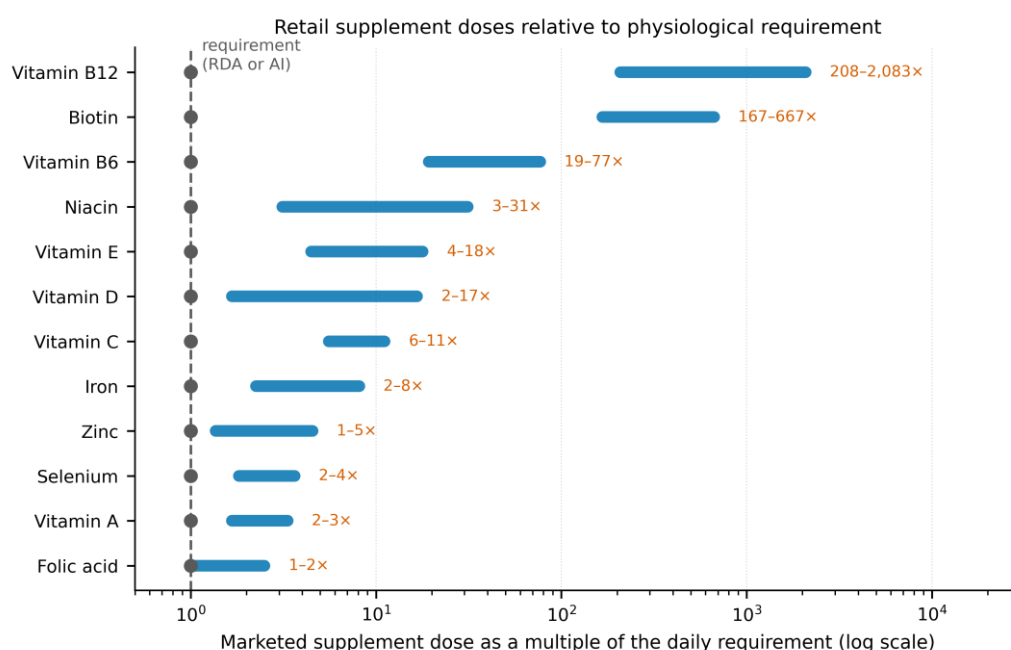


Figure 3. Marketed supplement doses expressed as multiples of the daily requirement, logarithmic scale. Bars span typical retail dose ranges; the dashed line marks the recommended dietary allowance or adequate intake. For several nutrients the marketed dose exceeds requirement by two to three orders of magnitude. Requirement values are adult reference intakes; retail ranges are illustrative of widely available products and should be confirmed against a systematic label survey.

Two features of this distribution stand out. It is not a matter of small overshoots: for the nutrients at the top of the figure, a single tablet delivers more than a year of dietary requirement. And the multiples are largest precisely for the nutrients conventionally regarded as harmless, which is not a coincidence — it is the safety assumption that permits the dose.

5.2 Chronic exposure is the relevant exposure

The reassurance usually offered about water-soluble vitamins derives from single-dose pharmacokinetic studies. Those studies are sound, and their finding is real: absorption of several water-soluble vitamins saturates, and surplus appears in urine. Vitamin C is completely absorbed at a single 200 mg dose, with bioavailability declining above 500 mg and the absorbed excess excreted, from which the authors concluded that daily doses above 400 mg have no evident value.^{48, 49} Vitamin B12 absorption via intrinsic factor saturates at roughly 1.5–2 µg per meal, so that only about one per cent of a large oral dose crosses by passive diffusion.⁵⁰

But the exposure of interest is not a single dose. It is 500 µg taken every morning for three months, or three years. Chronic high intake is a different pharmacological situation, and there is mechanistic reason to expect adaptation, since indefinite unchanged handling is unlikely: high ascorbate concentrations down-regulate the intestinal transporter SVCT1, reducing uptake substantially in cell models, and the transporters for thiamin, riboflavin, biotin and pantothenate are likewise substrate-regulated.^{51–53} Direct human data on transporter adaptation under sustained megadosing are lacking; this is a mechanistically supported expectation, not a demonstrated clinical phenomenon. The narrower and safer claim is simply that acute pharmacokinetics do not license confident inference about chronic exposure at two or three orders of magnitude above requirement, and that reassurance drawn from the former is being applied to the latter without warrant.

5.3 Who actually exceeds the limits

Population survey data identify the route. Roughly 58% of United States adults report using a dietary supplement in the preceding thirty days, rising to about 80% of women aged sixty and over; industry surveys report a higher figure of around 74%, a difference attributable to non-probability online panels and a broader definition of use, and not to any real disagreement about behaviour.^{54, 55} Among adults, adding supplements to food raises the proportion exceeding an upper limit for calcium, iron, selenium, zinc, vitamin A, folic acid, vitamin B6, vitamin C and vitamin D. For most nutrients and most age groups the proportion affected remains below five per cent.⁵⁶ It is not uniformly small, however, and the exceptions matter: in an earlier survey cycle 10.3% of the population aged two and over exceeded the limit for niacin and 8.4% for zinc once fortification and supplements were counted,⁴¹ and 9.0% of adults aged 51 to 70 exceeded the calcium limit.⁵⁶ Among supplement users specifically, roughly one in ten exceeded the limits for iron and for zinc.⁵⁷

In children the figures are of a different order. Among supplement-using children aged two to eight years, 84% exceed the upper limit for zinc, against 45% of non-users, with 72% above the vitamin A limit and 71% above that for folic acid.⁴⁰ Canadian survey data show a similar pattern, with more than 80% of children exceeding the limits for vitamin A and niacin in some age groups.⁵⁸ One caveat: the paediatric zinc limit is extrapolated from adult data and its adverse-effect basis is contested, so these proportions indicate exposure above a stipulated threshold, not demonstrated harm.

One limitation cuts against the reassurance the small per-nutrient percentages appear to offer. Every published analysis reports exceedance one nutrient at a time. We could find no estimate, for any general population, of the proportion of people exceeding the limit for at least one nutrient considered across the whole set, even though the corresponding aggregation has been performed on the deficiency side.³⁰ The distinction is not trivial. What can be said without further data is only that the aggregate is bounded below by the largest single-nutrient rate and above by the sum of the individual rates; independence would place it near the upper bound and clustering among the same multivitamin-using individuals pulls it toward the lower, and the interval between them is wide. We do not attempt to narrow it. The analysis that would be neither novel nor expensive: apply the usual-intake methods already used for single nutrients to the National Health and Nutrition Examination Survey dietary recalls combined with its Dietary Supplement Database, and report the proportion of participants whose usual intake exceeds the upper limit for at least one nutrient in a prespecified set. Until that is done, the per-nutrient framing cannot be taken to bound population-level over-exposure.

Product composition compounds this. An analytical survey of adult multivitamin and mineral products found niacin labelled above its upper limit in 31% of products, with smaller proportions above the limit for magnesium, manganese, zinc, vitamin B6 and vitamin C.^{59, 60} Measured content also departs from label: the same programme found most ingredients present in excess of the declared amount, by around a quarter for selenium and iodine.

5.4 Dose multiples do not by themselves demonstrate harm

It would be convenient to reason directly from these multiples to harm, but for several of the most extravagantly dosed nutrients the evidence of harm is thin or absent. Vitamin B12 has no upper limit precisely because no adverse effect has been demonstrated, and there is no controlled evidence that a daily intake two hundred times the requirement produces symptomatic harm in a healthy adult.^{15, 50} The one reproducible effect attributed to high-dose vitamin B12 alone is an acneiform eruption, described in case reports and a small case series

and for which a mechanism has been demonstrated experimentally, and it is uncommon.^{61–63} Where trials of high-dose B vitamins have reported excess cancer or accelerated renal decline, the preparations combined folic acid with vitamin B12 and sometimes vitamin B6, and the signal is generally attributed to folic acid; these designs cannot isolate a single agent.^{64–66} Observational associations between high circulating vitamin B12 and cancer are, on the evidence, largely a matter of reverse causation, since hepatic disease and malignancy themselves raise measured concentrations.⁶⁷

The defensible claim is narrower than the dose multiples might invite, and we state it deliberately. It is not that a two-hundred-fold multiple of requirement is known to be harmful. It is that such an exposure, sustained for years, is one about which the long-term evidence is largely absent in either direction, and that the absence of an upper limit — a statement about what has been studied — has been widely read as a statement about what is safe. That inferential slide is the problem, and it is a problem an assessment instrument could at least make visible by recording the intake.

5.5 Water solubility does not predict safety

Underlying the whole pattern is a heuristic that deserves to be named and retired: that water-soluble vitamins are inherently safe because surplus is excreted. It is memorable, it is taught, and it is wrong often enough to be dangerous.

Vitamin B6 is the decisive case. It is freely water-soluble, and it causes a dose- and duration-dependent sensory neuropathy, classically described at multi-gram intakes but reported at a few hundred milligrams daily and, in recent regulatory review, without a clearly identifiable threshold.^{5, 68} The European Food Safety Authority reduced its upper limit to 12 mg per day in 2023.⁶⁸ Australia's Therapeutic Goods Administration has required a peripheral-neuropathy warning on products supplying more than 10 mg daily since 2022, and in November 2025 rescheduled oral preparations supplying more than 50 mg but not more than 200 mg per recommended daily dose as Pharmacist Only medicines with effect from 1 June 2027; preparations above 200 mg daily were already prescription-only.^{69, 70} Niacin, also water-soluble and essentially completely absorbed even at gram doses, causes flushing at intakes barely above requirement and hepatotoxicity at pharmacological doses.⁵⁰ Chronic high-dose vitamin C produces dose-dependent hyperoxaluria and is associated with kidney stones in men.⁷¹ Folic acid at fortification-plus-supplement intakes leaves unmetabolised folic acid in circulation and can mask the haematological presentation of B12 deficiency, although its supposed acceleration of the accompanying neuropathy is contested.^{72, 73} Biotin causes no toxicity in the conventional sense yet produces a clinically serious harm of a different kind, interfering with streptavidin-based immunoassays including cardiac troponin; the regulator has received one report of a death following a falsely low troponin result, although causation in that case was not established.⁷⁴

Set against this, the fat-soluble vitamin K has no upper limit because no adverse effect has been demonstrated for phyloquinone or the menaquinones — while the synthetic, water-soluble vitamer menadione is hepatotoxic and has been withdrawn from human use.²⁴ The solubility rule does not merely fail at the margins; for vitamin K it inverts. The point is not new: these arguments were set out in a review published four decades ago.⁷⁵ They have not been absorbed into how nutrient intake is assessed.

6. HOW EXISTING INDICES TREAT THE UPPER SIDE

The monotone construction of diet-quality indices is not an error but an inheritance, and understanding its origin clarifies where it does and does not still apply. The first generation of

indices was built in the 1990s to summarise adherence to dietary guidance in populations where the dominant nutritional problems were inadequate intake of protective foods and excessive intake of a small number of constituents (sodium, saturated fat, added sugar) for which the direction of harm was unambiguous.^{9,10} The scoring reflected that problem structure faithfully. Where a constituent was harmful in a known direction, it was reverse-scored; where a food group was beneficial, more of it earned more points up to a standard.

Two things have changed since. Fortification and supplementation have made high micronutrient intakes common in high-income populations, where they were once exceptional, and the upper tail of the intake distribution is now populated by people who are neither ill nor unusual.^{56,57,76} And the indices themselves have migrated from their original role in population surveillance into individual-facing applications, where the same score is presented to a person as an account of their own diet, a migration visible in composite health scores such as Life's Essential 8, whose diet component is drawn directly from this tradition,⁷⁷ and in nutrient-profiling systems built to rank individual foods.⁷⁸ Neither change invalidates the instruments for their original purpose. Both change what happens when the instruments are pointed at an individual whose intake sits at the top of the distribution, not the bottom.

We examined the scoring construction of eight widely used diet-quality instruments against their defining publications (Table 3, Figure 4). The pattern is consistent, and more nuanced than a simple charge of one-sidedness would suggest.

Index	Treatment of shortfall	Treatment of excess
HEI-2020 ^{79, 80}	Adequacy components scored up to a maximum standard	Moderation components (refined grains, sodium, added sugars, saturated fat) scored in reverse; no nutrient-level upper penalty
AHEI-2010 ⁸¹	Component intakes scored 0–10 toward a target	Trans fat, sugar-sweetened beverages and red/processed meat scored in reverse; capped rather than penalised beyond the cap
DASH score ⁸²	Higher intakes of favoured groups scored higher	Sodium, red meat and sweets scored in reverse by quintile
Mediterranean score ⁸³	Above-median intake of beneficial components scored 1	Below-median intake of meat and dairy scored 1; dichotomous, no upper penalty
Global Diet Quality Score ⁸⁴	Healthy food groups scored by intake category	Unhealthy groups scored in reverse; red meat and high-fat dairy carry a quantity-dependent penalty
Planetary Health Diet Index ⁸⁵	Adequacy relative to reference intakes	Exceeding reference intakes of animal-source foods scored down
NOVA-based metrics ⁸⁶	Not scored	Ultra-processed share scored in reverse; processing-based, not nutrient-based
Dietary diversity (MDD-W) ⁸⁷	Count of food groups consumed	Not scored

Table 3 Treatment of shortfall and excess in eight widely used food-based instruments. Nutrient-based indices, which behave differently, are considered separately in the text. Several penalise excess at the food-group level; none applies a two-sided rule at the level of an individual nutrient.

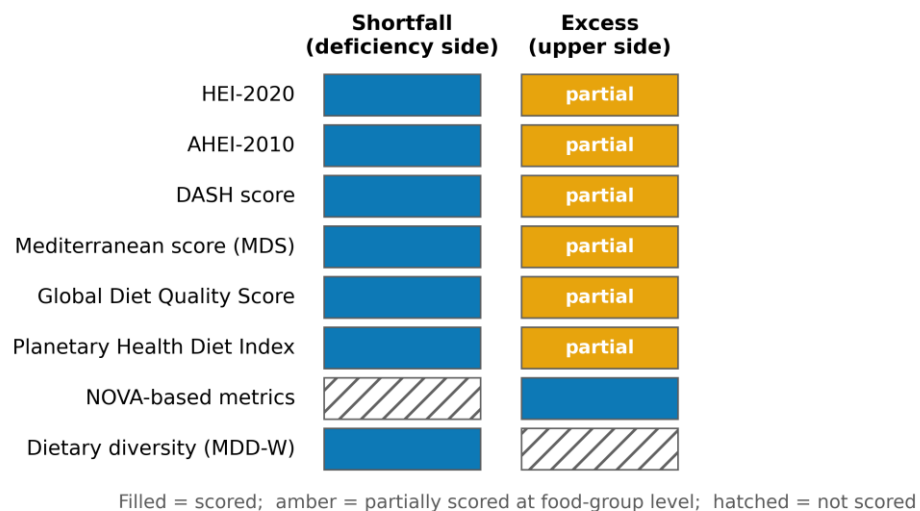


Figure 4. Treatment of shortfall and excess in eight food-based diet-quality instruments. Most instruments penalise excess for a limited set of food groups or dietary constituents, shown in amber; none applies a two-sided rule at the level of an individual micronutrient.

Two conclusions follow, and the second qualifies the first. Every index in the table other than the diversity indicator does penalise something on the upper side, and it would be inaccurate to describe them as purely monotone. HEI-2020’s moderation components, the reverse-scored elements of AHEI and DASH, and the animal-source penalties in the Global Diet Quality Score and the Planetary Health Diet Index are all mechanisms for scoring down an intake that has become too high.^{79, 81–85} Any claim that diet-quality indices ignore excess altogether is wrong, and we do not make it.

There is a second and more consequential omission, which follows directly from the previous section. None of these instruments counts supplements. This is not an oversight but a design decision, stated explicitly for the Healthy Eating Index, which assesses diet in concordance with the dietary guidelines’ focus on meeting nutrient needs from foods and beverages rather than from supplements.⁷⁹ The same food-only construction applies to the others, and to dietary diversity indicators, which count food groups by definition. The consequence is structural: because supplements are the route by which individuals reach intakes above upper limits, an instrument that excludes them cannot detect excess even in principle. It can measure dietary pattern quality and it can measure adequacy; it is blind to the tail of the distribution where the second cliff lies. Food-group diversity indicators share the same structural limitation for the same reason: a supplement is not a food group, so no count of groups can register it.

What none of them does is apply a two-sided rule at the level of the nutrient. The moderation components operate on food groups and dietary constituents chosen for their association with chronic disease (sodium, added sugars, saturated fat, processed meat), not on micronutrients with defined upper limits. An intake of preformed vitamin A, selenium or iodine well above the UL is invisible to every instrument in the table, because none of them scores those nutrients at all, in either direction. The gap is therefore narrower than it first appears, and more specific: it lies at the micronutrient level, where upper limits exist and are not consulted.

That holds for the food-based instruments in Table 3, which are the ones generating the published diet-quality literature. It does not hold for the whole field, and the proposal below should be read against a smaller nutrient-based literature; it does not fill an empty space. The

PANDiet index computes a probability of adequacy for each nutrient and adds a moderation sub-score in which a nutrient whose average intake exceeds its upper limit is scored zero; its authors introduced that sub-score precisely because the mean adequacy ratio and the mean probability of adequacy take no account of upper levels of intake.⁸⁸ A nutritional quality index developed for sportspeople scores a nutrient at its maximum only when intake lies between two-thirds of the recommended intake and the upper limit, scores it zero above the limit, and computes a second score including supplements.⁸⁹ My Nutrition Index scores each nutrient continuously, highest within the recommended range and lower with increasing distance from it in either direction.⁹⁰ The Total Nutrient Index was built to capture usual intake from all sources, supplements included, for a set of under-consumed micronutrients, and reports a food-and-beverage score alongside a total score.^{91, 92}

The argument of this review is therefore not that the problem is unrecognised but that its treatment has stayed partial and confined to a small literature. Each of these instruments takes a step the routine indices do not, and none takes all of them together. Where an upper limit is used it functions as a threshold, the score falling to zero once intake crosses it, and does not decline across a graded interval; where a lower threshold is stated explicitly it is a recommended intake rather than the median requirement, which carries forward the inferential problem set out in Section 3; supplements, where counted, are added into a single total, not separated from enrichment and fortification; and the depth of any penalty is not scaled to the strength of the evidence behind the limit. None of these indices has entered routine surveillance use.

7. IMPLICATIONS FOR INDEX CONSTRUCTION AND FOR PRACTICE

We propose four changes, ordered from least to most demanding.

State the reference value and the inference together. Any instrument that judges an individual's intake should name the reference value it uses and the conclusion that value supports. Where the EAR exists, it should be the comparator, and the output should be expressed as a probability of inadequacy and not as a pass or fail. Where only an AI exists, the instrument should say that intakes below it are uninformative. This costs nothing but candour.

Count supplements. This is the single change that would do most, because it is the channel through which excess occurs. An instrument that omits supplements is not merely incomplete; it is systematically unable to observe the phenomenon a bidirectional score exists to detect. Total usual intake from food and supplements can be modelled with established survey methods, and any index intending to speak about the upper end must do so. Supplements must also be separated from enrichment and fortification instead of being folded into a single total, because several limits are defined for a particular chemical form or source and an undifferentiated total cannot be compared against them.

Score bidirectionally where the evidence allows. For the subset of nutrients with a defined limit and a documented upper limb (iodine, selenium, preformed vitamin A, iron, zinc, vitamin B6 and a small number of others), a two-sided rule is constructible today, with penalties weighted by the evidentiary strength of the limit, not applied uniformly.

Where no official limit exists, use an established derivation rather than an invented rule. It is tempting to fill the gap with a heuristic such as a fixed multiple of the requirement, and we considered doing so. It does not survive scrutiny. Absorptive and excretory capacity are governed by transporter kinetics and plasma concentration, which bear no necessary relationship to a nutrient's dietary requirement, so a requirement-multiple is dimensionally

arbitrary; deriving maximums from the recommended intake has been criticised on precisely these grounds.⁹³ A defensible alternative already exists. The United Kingdom's Expert Group on Vitamins and Minerals derived Guidance Levels from the highest intake at which no adverse effect has been observed, with an explicit uncertainty factor, for exactly those nutrients lacking a formal limit, and the observed-safe-level and highest-observed-intake methods provide the same service.^{93, 94} These yield citable numbers today and rest on human no-adverse-effect data in humans, and not on arithmetic. The next section sets out what an instrument built on these four principles would look like.

7.1 What would need to be shown

The proposal in this review is a design argument, not an empirical finding, and it should be tested as such, not adopted on the strength of its plausibility. A bidirectionally scored index could be built from existing national survey data in which both dietary intake and supplement use are recorded, and compared against conventional monotone scoring. Section 8.7 sets out the three stages at which that comparison would either acquire evidential support or fail to.

8. A PROPOSED TWO-SIDED COMPONENT SCORE

The preceding sections identified three properties that a nutrient assessment would need if it were to represent the upper limb of the intake–risk relationship: it must compare intake against a reference value that licenses the inference being drawn, it must count the sources through which high intakes are actually reached, and it must be able to score an intake down for being too high. No instrument in routine use has any of the three, and the nutrient-based indices described in Section 6 have some but not all. This section specifies a component score that would combine them. It is a design proposal derived from the evidence reviewed above; it has not been constructed, applied to survey data or validated against any outcome, and nothing here constitutes a recommendation for use.

Set against that literature, what is proposed here is a combination rather than a first. Its elements are a continuous function anchored on the estimated average requirement at the lower end and on the upper limit at the upper end, declining across a graded interval instead of switching at a threshold; application to every micronutrient carrying a reference value and not to a selected set; decomposition of intake into food, enrichment and fortification, and supplements; and a penalty depth scaled to the strength of the evidence behind each limit. We are not aware of a published index combining these, but for the reason given in Section 2 that records only what we located, not what exists.

8.1 Which nutrients it covers

The module applies to every micronutrient carrying a reference value, whether or not an upper limit has been set for it. In practice that is the set of vitamins, minerals and trace elements for which the National Academies, the United Kingdom expert committees or the European Food Safety Authority publish a requirement or an adequate intake. Table 4 lists twenty-six entries and not one per nutrient because thiamin, riboflavin and pantothenic acid share a row, the evidence position being identical for all three. Nutrients that carry a reference value but for which we located no excess-side outcome literature (molybdenum is one) do not appear in Table 4; under the module they would be scored on the deficiency side and recorded in Tier 4 on the upper side. Two considerations make the wider scope preferable to restricting it to nutrients with demonstrated harm.

The first is that documented effects of excess are not confined to a few nutrients. Assembling the excess-side literature for each micronutrient yields described adverse effects for the large

majority, from randomised evidence at one end to single clinical reports at the other (Table 4). Restricting the module to a shortlist would omit most of what is known.

The second is that a nutrient without an upper limit is not thereby a nutrient known to be safe. A limit is recorded as not determinable when adverse-effect data are insufficient, a distinction the National Academies state explicitly.²⁷ Excluding such nutrients would convert an evidence gap into a safety finding — the inference this review has criticised elsewhere. It would also fail where it is most needed: as Section 4 showed, the nutrients without limits are among those marketed at the highest multiples of requirement, so a module that skipped them would be silent precisely where exposure is most extreme.

The scope is therefore universal, and the strength of evidence is carried explicitly within the scoring, not used to decide what enters it. This does not contradict the earlier statement that any bidirectional scheme will be partial. Scoring is universal; penalties are not. Every micronutrient is scored and every intake recorded, but a penalty on the upper side is applied only where evidence supports one, and Tier 4 exists to carry the nutrients where it does not.

8.2 Anchors

The lower anchor is the estimated average requirement rather than the recommended dietary allowance, for the reason set out in Section 2: the recommended allowance is a population reference value and supports no inference about an individual, whereas comparison against the median requirement yields a probability that can be honestly reported. The upper anchor is the upper limit where one exists, and where none does, the Guidance Level derived by the United Kingdom expert group from the highest intake at which no adverse effect has been observed.^{93, 94} Because jurisdictions disagree (the selenium limit is 400 µg in the United States against 255 µg in the European Union, and the vitamin E limit 1,000 mg against 300 mg), an implementation must declare which system it uses and apply it consistently across all nutrients instead of selecting the more convenient value for each. We would use the reference values of the jurisdiction in which the assessment is made, since those are the values against which its dietary guidance is written; where a nutrient carries a limit in one system and not in another, the available value should be used and the asymmetry reported. For nutrients with neither, the module records intake but applies no upper-side penalty.

8.3 The scoring function

The function is piecewise and asymmetric, mirroring the asymmetry of the evidence on the two sides (Figure 5a). The score rises from zero at the lower anchor to a maximum, holds across a plateau of intakes at which neither deficiency nor excess is expected, and declines as intake approaches and passes the upper anchor.

Three parameters govern its shape, and each is a judgement rather than a datum: where the plateau begins, where it ends, and whether the penalty reaches zero at the upper anchor or continues to decline beyond it. We state them as choices requiring justification because the alternative, settling them silently, is the practice this review criticises. Different defensible choices will produce different scores for the same diet, and any application should report how sensitive its results are to them.

Written out, and with x denoting usual intake from all sources, E the lower anchor (the estimated average requirement), R the recommended intake, U the upper anchor, p_1 the plateau onset, p_2 the penalty onset and $d \in [0,1]$ the tier-specific penalty depth, the component score $S(x)$ on a 0–100 scale is:

$$S(x) = 0 \quad \text{for } x < E$$

$$\begin{aligned}
S(x) &= 100 (x - E) / (p_1 - E) && \text{for } E \leq x < p_1 \\
S(x) &= 100 && \text{for } p_1 \leq x < p_2 \\
S(x) &= 100 [1 - d (x - p_2) / (U - p_2)] && \text{for } p_2 \leq x < U \\
S(x) &= 100 (1 - d) && \text{for } x \geq U
\end{aligned}$$

The penalty depth d takes the value 1 in Tier 1, 0.6 in Tier 2, 0.25 in Tier 3 and 0 in Tier 4, so that a Tier 1 nutrient scores zero at the upper anchor while a Tier 4 nutrient is never penalised. For illustration we would set $p_1 = R$ and $p_2 = R + 0.6 (U - R)$, placing the plateau between the recommended intake and a point three-fifths of the way to the limit; Figure 5 is drawn with these values. Both are stipulations rather than estimates, and an application adopting different ones should say so and report how far its results move.

8.4 Four evidence tiers

Upper limits are not equally well founded. They are derived by dividing an observed adverse-effect level by an uncertainty factor that has ranged from 1 to 36 across nutrients, sometimes on the basis of a single study or of animal data, and jurisdictions differ by up to three-fold on the same nutrient.^{8,27–29} Applying an identical penalty above every limit would treat these as equivalent. We therefore propose that the depth of the upper-side penalty be scaled to the evidence behind the limit, in four tiers defined as follows.

Tier 1. Harm demonstrated in humans by systematic review, meta-analysis or randomised trial. The penalty is applied in full: the component score falls to zero at the upper anchor.

Tier 2. Harm documented in clinical series, cohort studies, or the derivation of the upper limit itself. The penalty is attenuated: the score falls to an intermediate value and not to zero.

Tier 3. Harm described in case reports, animal studies or authoritative clinical summaries, without dose–response data. The penalty is shallow and functions as a flag rather than a sanction.

Tier 4. No adverse effect identified at any studied intake. No penalty is applied. The intake is nonetheless recorded and reported as a multiple of requirement, accompanied by the statement that the upper end of the range has not been characterised. This is a weaker claim than a penalty and a stronger one than silence, and it is what the evidence supports. It should also be said plainly that this is where the module is weakest, and that its weakness falls exactly where Section 5 located the most extreme exposures. The nutrients marketed at the largest multiples of requirement — vitamin B12 and biotin above all — are among those for which no limit has been set, so no penalty can be justified and none is applied. Recording the intake and stating the gap is the most the evidence permits; it is less than the situation warrants.

One distinction should be made explicit, because the tiers and the scope of the proposal answer different questions. A tier records how strong the evidence is that harm occurs at high intake; it is not a decision to include the nutrient in an operating instrument. Sodium and vitamin D are placed in Tier 1 because trial and meta-analytic evidence of upper-end harm exists for both, but for the reason given in Section 4.7 we would exclude both, and alcohol, from any implementation until the shape of their intake–risk relationships is less contested. Table 4 grades evidence; it does not settle scope.

Figure 5b shows the four penalty profiles. Table 4 assigns each micronutrient to a tier on the strongest study design located for it.

Nutrient	Tier	Documented effect of excess	Strongest design
Vitamin B6	1	Sensory peripheral neuropathy	Systematic review ⁹⁵
Preformed vitamin A	1	Teratogenicity; hepatotoxicity; reduced bone density	Meta-analysis (fracture); cohort (teratogenicity) ^{96, 97}
Beta-carotene	1	Increased lung cancer and mortality in smokers	Two randomised trials ^{21, 22}
Vitamin D	1	Hypercalcaemia; nephrocalcinosis; more falls at megadose	Network meta-analysis ⁹⁸ ; case report ⁹⁹
Manganese	1	Manganism; impaired cognition	Meta-analysis ¹⁰⁰ ; review ¹⁰¹
Sodium	1	Raised blood pressure; renal and vascular injury	Cochrane meta-analysis ¹⁰²
Selenium	1	Selenosis: hair and nail loss, neuropathy, GI upset, mood change	Randomised trial ⁴ ; population studies and reviews ¹⁰³
Vitamin E	1	Haemorrhagic tendency; raised pneumonia risk in smokers	Meta-analysis of supplementation trials ²³ ; cohort analysis ¹⁰⁴
Zinc	2	Copper deficiency: anaemia, myeloneuropathy	Case series ³¹ ; case report ¹⁰⁵
Iron	2	GI effects; organ damage from overload	Limit derivation ²⁴
Vitamin C	2	Hyperoxaluria; oxalate nephropathy; osmotic diarrhoea	Case report ¹⁰⁶ ; limit derivation ²⁶
Iodine	2	Hyper- and hypothyroidism; autoimmune thyroiditis	Population studies across the intake range ^{1, 2} ; clinical review ¹⁰⁷
Fluoride	2	Dental and skeletal fluorosis	Limit derivation ²⁵
Calcium	2	Reduced iron and zinc absorption; stone risk from supplements	Reviews ²⁹
Folic acid	2	Masks the haematological presentation of B12 deficiency	Review ³²
Niacin	2	Flushing; dose-dependent hepatotoxicity	Clinical; limit derivation ⁵⁰
Vitamin B12	3	Acneiform eruption; a reported toxicity syndrome	Case reports ^{62, 108}
Chromium	3	Gastric irritation and ulceration at high dose	Animal study ¹⁰⁹
Magnesium	3	Osmotic diarrhoea; hypermagnesaemia in renal impairment	Clinical; limit derivation ²⁹
Copper	3	Gastroenteritis; hepatic injury in accumulation states	Clinical description ²⁴

Nutrient	Tier	Documented effect of excess	Strongest design
Phosphorus	3	Hyperphosphataemia; vascular and bone effects in kidney disease	Review ¹¹⁰
Potassium	3	Hyperkalaemia and arrhythmia, chiefly in renal impairment	Clinical review ¹¹¹
Choline	3	Hypotension; fishy body odour	Limit derivation ⁵⁰
Biotin	3	No toxicity; interferes with clinical immunoassays	Regulatory safety communication ⁷⁴
Vitamin K	4	None for K1/K2; menadione is hepatotoxic; antagonises warfarin	Clinical description ²⁴
Thiamin, riboflavin, pantothenic acid	4	No adverse effect identified at any studied oral intake	Absence of evidence ⁵⁰

Table 4 Assignment of micronutrients to the four evidence tiers, with the documented effect of excess and the strongest study design located for each. Tier assignment reflects the strongest available design, not a systematic appraisal of all excess-side evidence, and should be revisited as upper limits are revised. Tier 4 records that the question has not been investigated, not that safety has been established. Sodium and vitamin D are graded here but excluded from the proposed module for the reason given in Section 4.7. The mapping is author-constructed and was not produced by duplicate independent screening; it should be read as a structured account of the authors' reading and not as a validated grading.

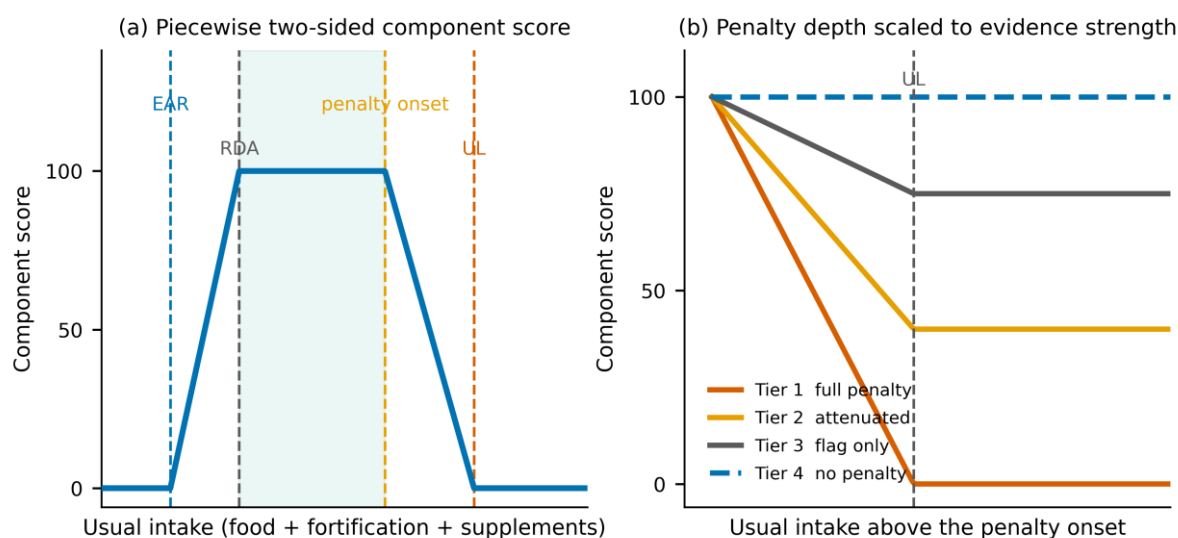


Figure 5. The proposed component score. (a) A piecewise function rising from the estimated average requirement to a plateau, then declining toward and beyond the upper anchor. Its three shape parameters (plateau onset, penalty onset, and behaviour beyond the anchor) are design choices requiring justification, not data. (b) Penalty depth scaled to the evidence behind the upper limit, in the four tiers defined above: full penalty, attenuated, flag only, and no penalty. Axes are illustrative: this is a schematic of the proposed form and not a fitted function.

8.5 Intake must be counted by source

The module requires usual intake disaggregated into naturally occurring food, enrichment and fortification, and supplements. This is a requirement rather than a refinement, for two reasons established earlier. Several limits are defined for a particular chemical form or source: magnesium for supplemental intake only, folate for synthetic folic acid, vitamin A for preformed retinol, niacin and vitamin E for synthetic or supplemental forms. An undifferentiated total cannot be compared against them without error.^{28, 29, 31} And the sources

differ in which population they carry past a limit, supplements predominating among adults and fortification among young children.⁶¹ No existing diet-quality instrument disaggregates in this way, and none counts supplements at all.

8.6 Two outputs, not one

The module should report two quantities instead of collapsing them. The first is an adequacy statement referenced to the estimated average requirement and expressed as a probability. The second is a separate excess flag, referenced to the upper anchor and tiered as above. Combining them into a single number would reproduce in miniature the problem this review identifies: a person consuming twice the requirement of one nutrient and a tenth of another is not well described by an average of the two, and the two findings call for different responses. Reporting them separately also keeps visible the distinction between an inference about inadequacy, which the reference values support probabilistically, and an observation of intake above a stipulated threshold, which is a statement about exposure rather than about harm.

One consequence of reporting an excess flag needs stating, because it bears directly on clinical use. High intakes are frequently deliberate and correct: iron for diagnosed deficiency, folic acid in pregnancy, vitamin B12 in pernicious anaemia, vitamin D under supervision for documented insufficiency. Each of these would raise the flag. The flag is a statement about exposure relative to a population threshold and carries no judgement of clinical appropriateness, and any instrument implementing the module should say so in its output instead of leaving the inference to the reader. An implementation intended for clinical settings would also need an exemption path in which a recorded indication suppresses the penalty while preserving the intake record, so that a therapeutic regimen is not scored as a dietary fault. Pregnancy is the sharpest instance. The requirement for folate rises from 400 to 600 µg dietary folate equivalents and that for iron from 18 to 27 mg, while the corresponding upper limits, 1,000 µg of folic acid and 45 mg of iron, are unchanged,^{24, 50} so the interval between the two anchors narrows precisely where guidance directs intake upward. Periconceptional folic acid and iron supplementation are among the most widely recommended interventions in nutrition, and a woman following that guidance would be flagged on both nutrients. In this population an exemption path is not a refinement but a precondition of the module being usable at all.

8.7 What would establish whether this is worth building

The proposal is testable in three stages of increasing cost. The first is descriptive and could be undertaken with existing national survey data: apply the module to intakes recorded with supplement use and determine whether it identifies a distinguishable group whom conventional scoring rates favourably. The second is biochemical: establish whether that group differs in relevant biomarkers, which would indicate that the flag corresponds to something in the body and not only in the arithmetic. The third is prognostic and requires cohort data with long follow-up: determine whether two-sided scoring carries information about incident outcomes beyond conventional scoring. Only the third would justify adoption, and we would not advocate use of any such instrument before it.

9. LIMITATIONS

This review is narrative and purposive, not systematic, and no formal risk-of-bias appraisal was applied to the sources. Coverage of individual nutrients is illustrative. A comprehensive account of upper-limb evidence for every nutrient with a defined UL was beyond its scope, and the nutrients we discuss were selected partly because their upper limbs are well documented, which risks conveying more certainty about the excess side than the literature

supports. Our characterisation of each index rests on its defining publication and not on how it is applied in practice, and applications may differ. The argument is confined to micronutrients; macronutrient distribution and energy raise related but distinct issues that we have not addressed. It is also developed for non-pregnant adults and for children: pregnancy and lactation raise distinct questions: requirements that shift while limits do not, supplemental intakes that are deliberately elevated and usually correct, and outcomes arising in a second individual. We have not reviewed these, and they would require separate treatment before the module were applied in those groups. The scoring module of Section 8 is a theoretical proposal: it has not been constructed, applied to data or validated, and its parameters are stipulated, not estimated. The evidence tiers of Table 4 reflect the strongest design we located for each nutrient and not a systematic appraisal of all excess-side evidence, and tier assignment should be revisited as upper limits are revised; several rest on a single study, and Tier 4 records an absence of investigation rather than a demonstration of safety. Because assignment was made by the authors without duplicate independent screening, and because the tier determines the depth of the upper-side penalty, both the tiers and the penalty depths derived from them should be treated as a proposal requiring independent replication and not as a settled grading. Finally, the review is written from a high-income-setting perspective, and its emphasis on excess should not be read as suggesting that the upper limb deserves comparable weight in settings where inadequacy remains the dominant nutritional problem.

CONCLUSION

Nutrient–health relationships have two sides, but the instruments used to assess diet quality mostly see one. The gap is narrower than it first appears — several established indices already penalise excess at the food-group level — and it is also more specific: at the level of individual micronutrients, where upper limits have been defined and where the distance between the two cliffs is sometimes small, no widely used index looks upward at all, and the nutrient-based indices that do have not entered routine use. Alongside this sits a persistent inferential error in which population reference values are read as personal targets. Neither is remedied by a better slogan; each requires an explicit statement of what is being measured and what may be concluded from it. The evidence base will support a partial bidirectional index today, and it will support a fuller one only if the upper limb receives some fraction of the attention that has been given to the lower.

Author contributions

A.S.M. conceived the review, conducted the literature search and selection, and drafted the manuscript. A.H.H. contributed to the reference-value analysis and revised the manuscript critically. W.R. contributed to the analysis of supplement exposure and revised the manuscript critically. S.B. contributed the clinical interpretation and revised the manuscript critically. All authors approved the submitted version and agree to be accountable for the work.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Conflict of interest

A.S.M. is the author of a forthcoming general-audience book on lifestyle and health from which he receives royalties. The book refers to a companion web application that computes a

nutrition score; A.S.M. owns and developed that application, which is freely available and generates no revenue. The scoring module proposed in Section 8 was developed for this review and is not implemented in that application. A.H.H., W.R. and S.B. declare none.

Data availability

Data sharing is not applicable: no new data were created or analysed in this review.

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